

Ravikanth Gaddam Data Scientist

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SUMMARY

Data Scientist with over 2 years of experience in developing and deploying machine learning models, performing advanced data analysis, and leveraging data-driven insights to solve complex business problems. Strong expertise in Natural Language Processing (NLP), statistical modeling, and data visualization, with a proven track record of delivering impactful AI solutions across various domains including finance, healthcare, and customer service.

TECHNICAL SKILLS

Languages & Framework: Python, SQL, Scala, Bash, Shell Scripting, TensorFlow, PyTorch, Keras, scikit-learn, XGBoost, LightGBM, CatBoost, OpenCV

Database Technologies: MySQL, PostgreSQL, MongoDB, Cassandra, Redis, Apache Spark, Hadoop, Apache Kafka, Apache Flink, Hive, Pig, Snowflake, Redshift, BigQuery, Azure Synapse, Teradata

AI Models & Frameworks: NLTK, spaCy, Hugging Face Transformers, Gensim, BERT, GPT-3, T5, RoBERTa, LDA, Generative Adversarial Networks (GANs), DALL-E, Stable Diffusion, Reinforcement Learning, Zero-shot Learning, Few-shot Learning, FRCNN, YOLOv5, RCNN, LSTM, ARIMA, VGG16, ResNet50

Others: Docker, Kubernetes, Jenkins, Git, GitHub, GitLab, AWS SageMaker, Azure ML, Google Cloud AI, MLflow, TensorFlow Serving, ONNX

EXPERIENCE

Data Scientist, Capgemini

January 2024 – present | Remote, USA

- Designed, developed, and deployed advanced NLP solutions using state-of-the-art LLMs] such as GPT-3, BERT, T5, and GPT-Neo for applications like chatbots, document summarization, sentiment analysis, and question-answering systems.
- Built and fine-tuned Generative AI models for content creation tasks, such as automated report generation, product descriptions, and personalized customer support, leveraging transformers and diffusion models like DALL-E and Stable Diffusion.
- Conducted comprehensive data preprocessing for textual data, including tokenization, lemmatization, and stop-word removal, to prepare large volumes of unstructured data for model training and ensure high-quality inputs.
- Performed extensive model evaluation and optimization using metrics like F1-score, BLEU, ROUGE, perplexity, and MRR, and enhanced model performance using techniques such as Transfer Learning, RLHF, and knowledge distillation.
- Developed an intelligent customer support chatbot that reduced average handling time by 50% and increased customer satisfaction scores by 35%, leveraging GPT-4 for natural language generation and BERT for understanding context and sentiment.
- Created an automated document summarization system for legal and financial reports using T5 and BERT-based models, reducing document analysis time by 70% and increasing the accuracy of key information extraction by 45%.
- Collaborated with cross-functional teams to identify NLP use cases, gather requirements, and deliver tailored solutions, presenting findings and recommendations to stakeholders through detailed reports and visualizations.
- Continuously experimented with and implemented the latest advancements in NLP, LLMs, and Generative AI, such as zero-shot and few-shot learning, attention mechanisms, and transformer architectures, to enhance model capabilities.

Data Scientist, Bosch

Jun 2021 – Dec 2022 | Bengaluru, India

- Developed and deployed advanced machine learning models for predictive maintenance, autonomous driving assistance, and supply chain optimization, leveraging Python, TensorFlow, Keras, and scikit-learn.
- Led the development of deep learning models for computer vision tasks such as lane detection, obstacle detection, and pedestrian recognition using CNN, YOLOv5, and UNet, improving obstacle detection accuracy by 40%.
- Utilized cloud services like AWS SageMaker, Azure Databricks, and Google Cloud Platform (GCP) for scalable data processing, model training, and deployment, optimizing model performance and reducing training times by 50%.
- Designed and implemented various machine learning algorithms, including Time Series Forecasting, Anomaly Detection, and Classification models, achieving a 30% reduction in maintenance costs and a 25% increase in equipment uptime.
- Conducted hyperparameter tuning using grid search and random search techniques to optimize machine learning models for various business applications, resulting in a 15% reduction in false-positive rates for autonomous driving systems.
- Developed optimization algorithms (Linear Programming, Genetic Algorithms) and clustering techniques (K-Means, Agglomerative Clustering), reducing inventory holding costs by 20% and improving order fulfillment rates by 15%.
- Deployed machine learning models into production environments using Docker and Kubernetes, implementing CI/CD pipelines with Jenkins and GitLab CI for continuous integration, testing, and deployment, ensuring robust and scalable solutions.

EDUCATION

Stevens Institute of Technology

Master of Science; Major in Artificial Intelligence

Jan 2023 – May 2024

New Jersey, USA